

Unit 14

HYDROCARBONS

54 items · 141 marks

Syllabus sections in this unit

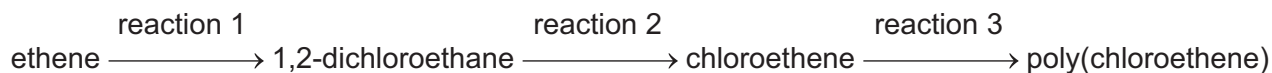
14.1 Alkanes

14.2 Alkenes

Unit 14: Hydrocarbons**1. M/J 26/P21/Q5(b)**

Context

- 5** Figure 5.1 describes a sequence of reactions starting with ethene to produce poly(chloroethene), PVC.

**Figure 5.1**

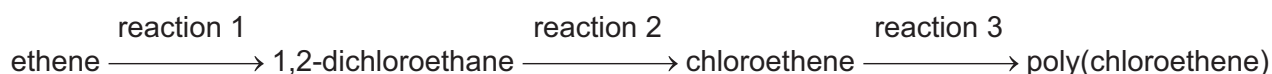
- (b)** In reaction 1, ethene reacts with $\text{Cl}_2(\text{g})$ at room temperature.
Name the mechanism for reaction 1.

[1]

2. M/J 26/P21/Q5(d)

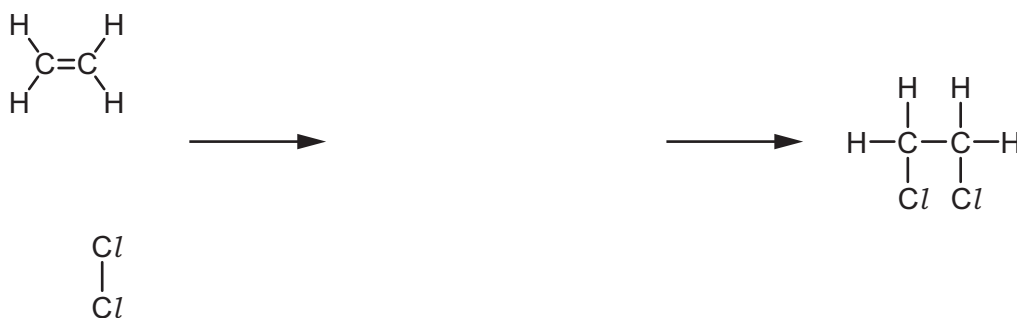
Context

- 5** Figure 5.1 describes a sequence of reactions starting with ethene to produce poly(chloroethene), PVC.

**Figure 5.1**

- (d)** Complete Figure 5.2 to show the mechanism for the reaction of ethene with Cl_2 in reaction 1.

Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.

**Figure 5.2**

[4]

3. M/J 26/P22/Q5(b)(iii)-(iv)

Context

- 5** Hexane, $\text{CH}_3(\text{CH}_2)_4\text{CH}_3$, reacts with $\text{Cl}_2(\text{g})$ in the presence of ultraviolet light to produce different organic products, including those shown in Table 5.1.

Table 5.1

structure of product	systematic name of product
$\text{CH}_3(\text{CH}_2)_5\text{Cl}$	
$\text{CH}_3(\text{CH}_2)_3\text{CHClCH}_3$	
$\text{CH}_3(\text{CH}_2)_3\text{CHClCH}_2\text{Cl}$	1,2-dichlorohexane
$\text{CH}_3(\text{CH}_2)_{10}\text{CH}_3$	dodecane
$\text{CH}_3(\text{CH}_2)_3\text{CH}(\text{CH}_3)\text{CH}(\text{CH}_3)(\text{CH}_2)_3\text{CH}_3$	5,6-dimethyldecane

Context: part (b) (introduction)

- (b)** The mechanism for the reaction of hexane with $\text{Cl}_2(\text{g})$ involves three stages.

stage 1 initiation

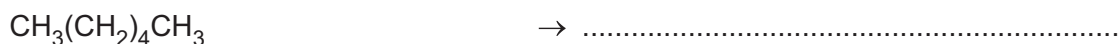
stage 2 propagation

stage 3 termination

In stage 1 and 2, free radicals are produced.

- (iii)** Suggest a reaction sequence of **two** propagation reactions that result in $\text{CH}_3(\text{CH}_2)_4\text{CH}_3$ forming $\text{CH}_3(\text{CH}_2)_3\text{CHClCH}_3$.

propagation reaction 1



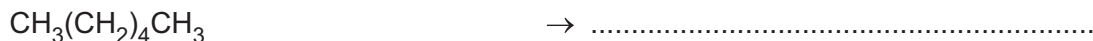
propagation reaction 2



[2]

- (iv)** Suggest a reaction sequence of **one** propagation reaction followed by a termination reaction that results in $\text{CH}_3(\text{CH}_2)_4\text{CH}_3$ forming $\text{CH}_3(\text{CH}_2)_{10}\text{CH}_3$.

propagation reaction



termination reaction



[2]

4. M/J 26/P22/Q5(c)

Context

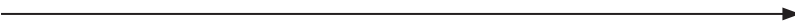
- 5** Hexane, $\text{CH}_3(\text{CH}_2)_4\text{CH}_3$, reacts with $\text{Cl}_2(\text{g})$ in the presence of ultraviolet light to produce different organic products, including those shown in Table 5.1.

Table 5.1

structure of product	systematic name of product
$\text{CH}_3(\text{CH}_2)_5\text{Cl}$	
$\text{CH}_3(\text{CH}_2)_3\text{CHClCH}_3$	
$\text{CH}_3(\text{CH}_2)_3\text{CHClCH}_2\text{Cl}$	1,2-dichlorohexane
$\text{CH}_3(\text{CH}_2)_{10}\text{CH}_3$	dodecane
$\text{CH}_3(\text{CH}_2)_3\text{CH}(\text{CH}_3)\text{CH}(\text{CH}_3)(\text{CH}_2)_3\text{CH}_3$	5,6-dimethyldecane

- (c) The proportion of each product varies depending on the type of free radical intermediate that is produced.

Table 5.2

primary free radical	secondary free radical	tertiary free radical
$\bullet\text{C}(\text{CH}_3)\text{H}_2$	$\bullet\text{C}(\text{CH}_3)_2\text{H}$	$\bullet\text{C}(\text{CH}_3)_3$
 increased stability of radical		

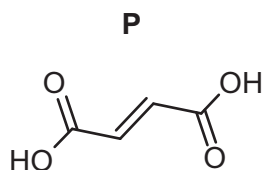
Use Table 5.2 to suggest which factor determines the increased stability of free radicals from primary to secondary to tertiary.

[1]

5. M/J 26/P22/Q6(b)(i)

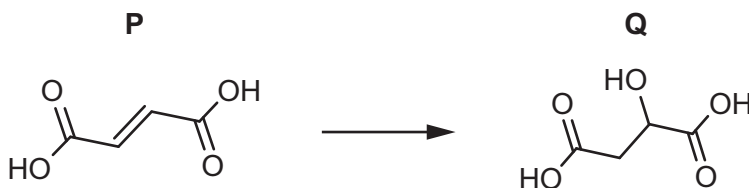
Context: part (a) (introduction)

- 6 (a)** Figure 6.1 shows the structure of compound **P**.

**Figure 6.1**

Context: part (b) (introduction)

- (b)** **P** reacts to produce compound **Q**.

**Figure 6.2**

- (i)** Identify the reagent and conditions required to convert **P** to **Q**.

reagent

conditions

[2]

6. M/J 26/P23/Q5(a)(ii)

Context

- 5** Propane, $\text{CH}_3\text{CH}_2\text{CH}_3$, reacts with $\text{Cl}_2(\text{g})$ in the presence of ultraviolet light to produce different organic products, including those shown in Table 5.1.

Table 5.1

structure of product	systematic name of product
$\text{CH}_3(\text{CH}_2)_2\text{Cl}$	1-chloropropane
$\text{CH}_3\text{CHClCH}_3$	2-chloropropane
$(\text{CH}_3)_2\text{CHCH}(\text{CH}_3)_2$	2,3-dimethylbutane
	hexane
	2-methylpentane

- (ii)** Name the mechanism for the reaction of propane and $\text{Cl}_2(\text{g})$ in the presence of ultraviolet light.

[1]

7. M/J 26/P23/Q5(b)(ii)-(iii)

Context

- 5** Propane, $\text{CH}_3\text{CH}_2\text{CH}_3$, reacts with $\text{Cl}_2(\text{g})$ in the presence of ultraviolet light to produce different organic products, including those shown in Table 5.1.

Table 5.1

structure of product	systematic name of product
$\text{CH}_3(\text{CH}_2)_2\text{Cl}$	1-chloropropane
$\text{CH}_3\text{CHClCH}_3$	2-chloropropane
$(\text{CH}_3)_2\text{CHCH}(\text{CH}_3)_2$	2,3-dimethylbutane
	hexane
	2-methylpentane

Context: part (b) (introduction)

- (b)** The mechanism for this reaction involves three stages.

stage 1 initiation

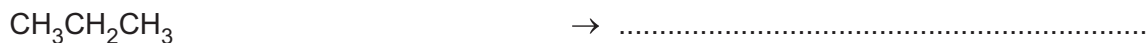
stage 2 propagation

stage 3 termination

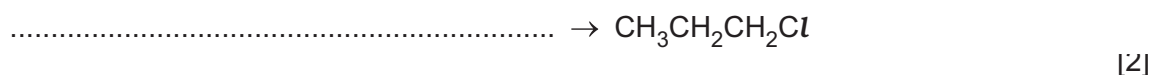
During the initiation reaction, a chlorine free radical is produced.

- (ii)** Suggest a reaction sequence of **two** propagation reactions that result in $\text{CH}_3\text{CH}_2\text{CH}_3$ forming $\text{CH}_3\text{CH}_2\text{CH}_2\text{Cl}$.

propagation reaction 1

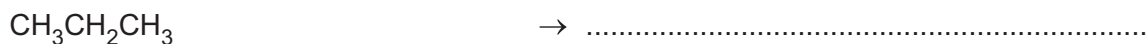


propagation reaction 2

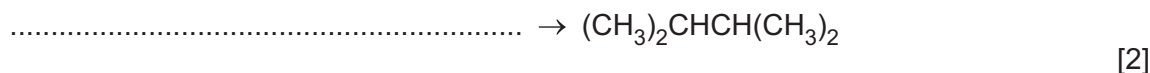


- (iii)** Suggest a reaction sequence of **one** propagation reaction followed by a termination reaction that results in $\text{CH}_3\text{CH}_2\text{CH}_3$ forming $(\text{CH}_3)_2\text{CHCH}(\text{CH}_3)_2$.

propagation reaction 3



termination reaction

**8. M/J 26/P24/Q5(a)**

Context

- 5** Long-chain alkanes from crude oil can be cracked to make alkenes.

- (a)** State the conditions used for the cracking of alkanes.

[1]

9. M/J 26/P24/Q5(b)(iii)

Context

5 Long-chain alkanes from crude oil can be cracked to make alkenes.

Context: part (b) (introduction)

(b) Propene is an alkene that is made by cracking. Alkenes are used in addition polymerisation.

(iii) Aqueous bromine is used to test for the presence of the C=C bond in propene.

State what is observed when aqueous bromine is mixed with propene.

[1]

10. M/J 26/P24/Q5(d)

Context

5 Long-chain alkanes from crude oil can be cracked to make alkenes.

(d) But-2-ene reacts with HBr(g). The HBr acts as an electrophile in the reaction.

(i) State how HBr acts as an electrophile in terms of electron transfer.

[1]

(ii) Complete Figure 5.1 to show the mechanism for the reaction of but-2-ene with HBr.

Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.



Figure 5.1

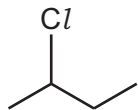
[4]

11. MAR 26/P22/Q3(c)

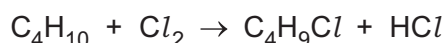
Context

- 3** Secondary halogenoalkanes are important building blocks in organic chemistry.

2-chlorobutane is an example of a secondary halogenoalkane.

2-chlorobutane**Figure 3.1**

- (c) Butane, C_4H_{10} , reacts with Cl_2 in the presence of ultraviolet light to form 2-chlorobutane, which has molecular formula C_4H_9Cl .



- (i) Name the mechanism of this reaction.

[1]

- (ii) Complete the reaction scheme given to show the formation of 2-chlorobutane.

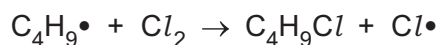
You may use molecular formulas in your answer.

initiation



propagation

propagation



termination

[2]

12. O/N 25/P21/Q3(c)(ii)-(iii)

Context

- 3** Cycloalkanes show similar chemical properties to alkanes but have the same empirical formula as alkenes.

Context: part (c) (introduction)

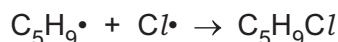
- (c) Cyclopentane reacts with Cl_2 in the presence of ultraviolet light to form C_5H_9Cl .

- (ii) Complete the equations to show the two propagation steps that follow the initiation step.



[2]

- (iii) The final step is shown.



Give the name for this step in the reaction.

[1]

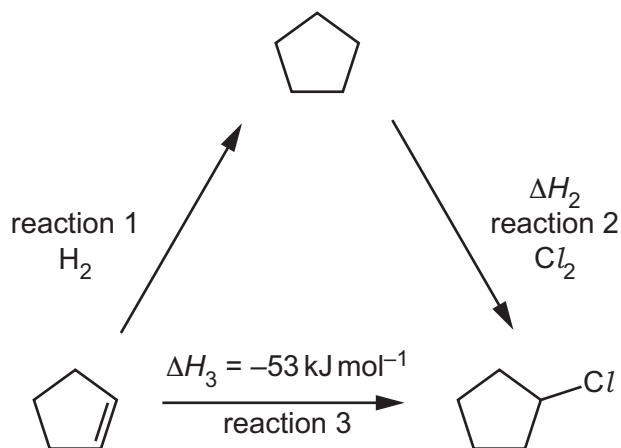
13. O/N 25/P21/Q3(d)(i)

Context

- 3** Cycloalkanes show similar chemical properties to alkanes but have the same empirical formula as alkenes.

Context: part (d) (introduction)

- (d)** Fig. 3.2 shows a reaction cycle involving cyclopentane, cyclopentene and C_5H_9Cl .

**Fig. 3.2**

- (i)** Identify a suitable reagent for reaction 3.

[1]

14. O/N 25/P21/Q3(e)(i)

Context

- 3** Cycloalkanes show similar chemical properties to alkanes but have the same empirical formula as alkenes.

Context: part (e) (introduction)

- (e)** Cyclopentene, C_5H_8 , reacts with hot concentrated acidified $KMnO_4$ to form compound **W**, $C_5H_8O_4$.

- (i)** Draw the structure of **W**.

[1]

15. O/N 25/P22/Q3(b)

Context

- 3** The alkanes are a homologous series of organic molecules. Alkanes are generally unreactive and are commonly used as fuels.

- (b)** Give two reasons to explain the general unreactivity of alkanes.

1

2

[2]

16. O/N 25/P22/Q3(c)

Context

- 3** The alkanes are a homologous series of organic molecules. Alkanes are generally unreactive and are commonly used as fuels.

(c) Alkanes with low relative molecular mass, M_r , are more useful than those found in heavier crude oil fractions.

Name the process that is used to obtain alkanes with low M_r from heavier crude oil fractions.

[1]

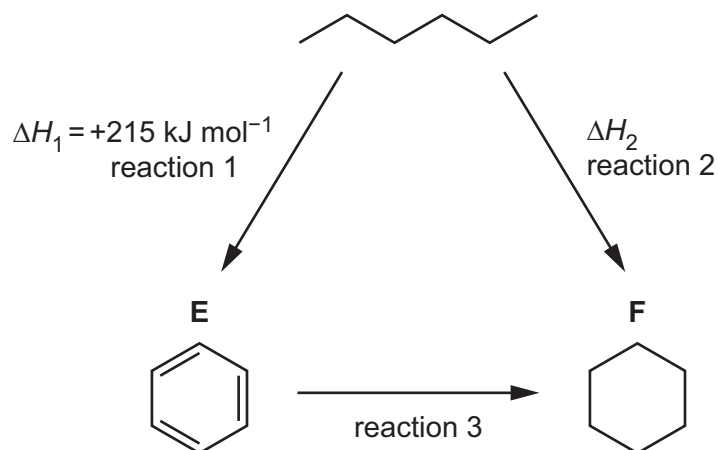
17. O/N 25/P22/Q3(e)(i)

Context

- 3** The alkanes are a homologous series of organic molecules. Alkanes are generally unreactive and are commonly used as fuels.

Context: part (e) (introduction)

(e) Hexane can be converted into compounds **E** and **F** at high temperature and pressure. Fig. 3.2 shows the reaction scheme involving hexane, **E** and **F**.

**Fig. 3.2**

(i) Identify a suitable reagent for reaction 3.

[1]

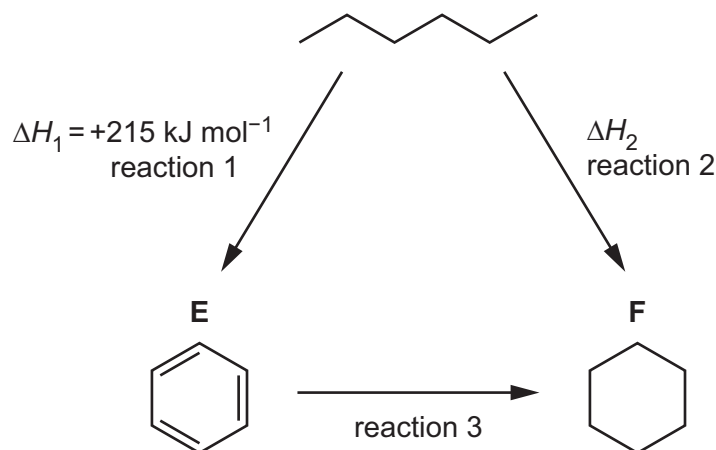
18. O/N 25/P22/Q3(e)(iii)

Context

- 3** The alkanes are a homologous series of organic molecules. Alkanes are generally unreactive and are commonly used as fuels.

Context: part (e) (introduction)

- (e)** Hexane can be converted into compounds **E** and **F** at high temperature and pressure. Fig. 3.2 shows the reaction scheme involving hexane, **E** and **F**.

**Fig. 3.2**

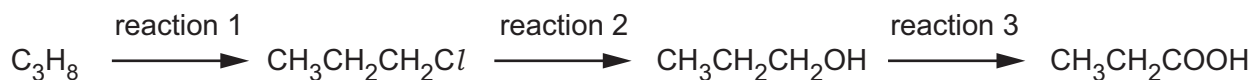
- (iii)** Write an equation for the complete combustion of hexane.

[1]

19. O/N 25/P22/Q4(a)(i)-(ii)

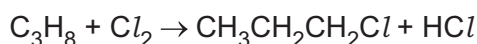
Context

- 4** Fig. 4.1 shows how propane, C_3H_8 , can be converted to propanoic acid, CH_3CH_2COOH .

**Fig. 4.1**

Context: part (a) (introduction)

- (a)** Reaction 1 in Fig. 4.1 takes place in the presence of sunlight.



The reaction takes place via initiation, propagation and termination steps.

- (i)** Name the mechanism shown by reaction 1.

[1]

- (ii)** Complete the mechanism for reaction 1.

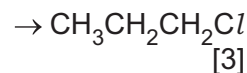
Construct equations to describe the steps of the mechanism.



propagation 1

propagation 2

termination



[3]

20. O/N 25/P23/Q3(c)(ii)-(iii)

Context

- 3** Cycloalkanes show similar chemical properties to alkanes but have the same empirical formula as alkenes.

Context: part (c) (introduction)

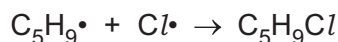
(c) Cyclopentane reacts with Cl_2 in the presence of ultraviolet light to form $\text{C}_5\text{H}_9\text{Cl}$.

- (ii) Complete the equations to show the two propagation steps that follow the initiation step.



[2]

- (iii) The final step is shown.



Give the name for this step in the reaction.

[1]

21. O/N 25/P23/Q3(d)(i)

Context

- 3** Cycloalkanes show similar chemical properties to alkanes but have the same empirical formula as alkenes.

Context: part (d) (introduction)

(d) Fig. 3.2 shows a reaction cycle involving cyclopentane, cyclopentene and $\text{C}_5\text{H}_9\text{Cl}$.

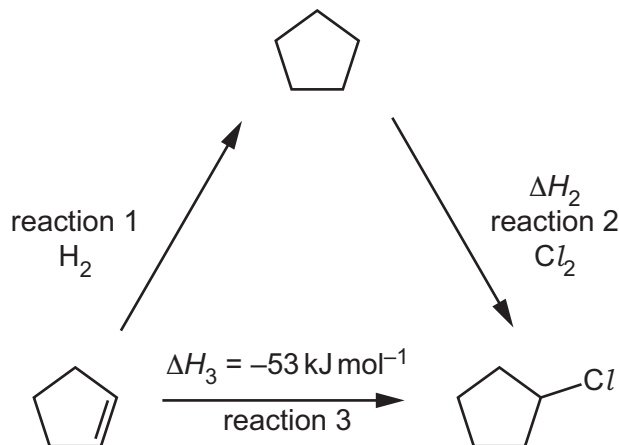


Fig. 3.2

- (i) Identify a suitable reagent for reaction 3.

[1]

22. O/N 25/P23/Q3(e)(i)

Context

- 3** Cycloalkanes show similar chemical properties to alkanes but have the same empirical formula as alkenes.

Context: part (e) (introduction)

- (e)** Cyclopentene, C_5H_8 , reacts with hot concentrated acidified $KMnO_4$ to form compound **W**, $C_5H_8O_4$.

- (i)** Draw the structure of **W**.

[1]

23. O/N 25/P24/Q4(a)

also Unit 13 (4 marks), Unit 22 (2 marks); this unit 5 marks

Context

- 4** Compounds **P** and **Q** are structural isomers.

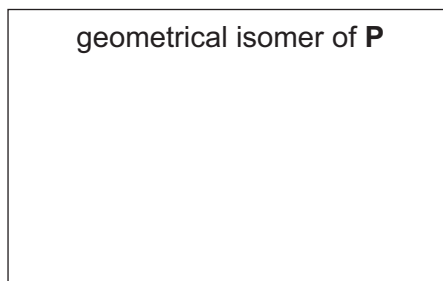
**Fig. 4.1**

- (a) (i)** Define structural isomerism.

[2]

- (ii)** **P** shows geometrical isomerism.

Draw the geometrical isomer of **P**. Explain why the two isomers are **not** identical.



explanation

[2]

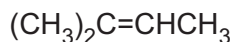
- (iii)** Both **P** and **Q** react with aqueous bromine.

Name the mechanism of this reaction.

[1]

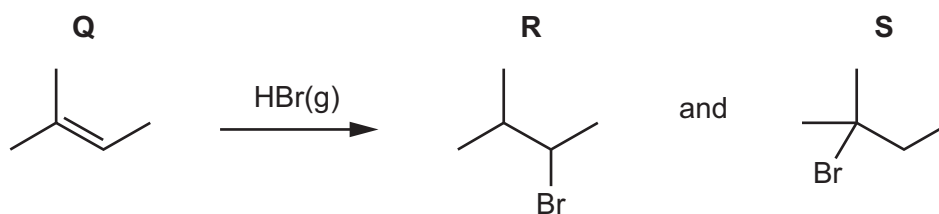
- (iv)** **Q** is oxidised by hot concentrated acidified $KMnO_4(aq)$, forming two different organic products.

Construct an equation for this reaction. Use [O] to represent an atom of oxygen from the oxidising agent.



[2]

- (v)** **Q** reacts with $HBr(g)$ to produce two structural isomers, **R** and **S**, as shown in Fig. 4.2.

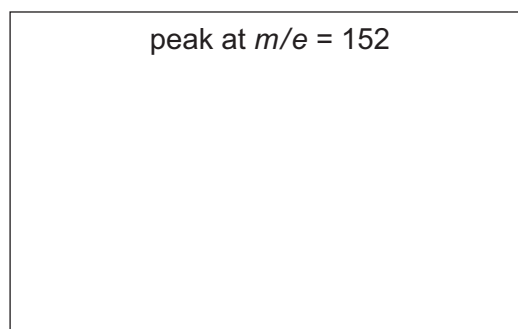
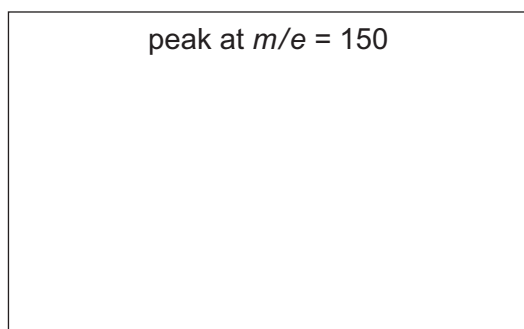
**Fig. 4.2**

State and explain why isomer **S** is the major product of the reaction.

[2]

(vi) The mass spectrum of **R** shows peaks at $m/e = 150$ and $m/e = 152$.

Suggest structures for the ions responsible for these peaks.



[2]

24. M/J 25/P21/Q4(b)(iv)

Context

4 The structure of vitamin C is shown in Fig. 4.1.

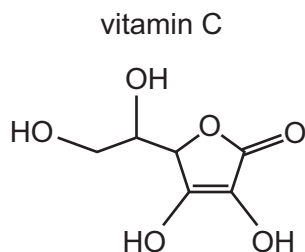


Fig. 4.1

Context: part (b) (introduction)

(b) The concentration of vitamin C is found by titration with $\text{I}_2(\text{aq})$.

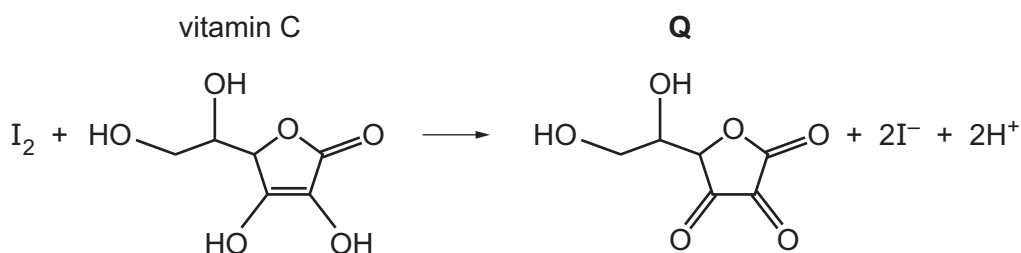


Fig. 4.2

A vitamin C tablet is dissolved in water to produce 200.0 cm³ of vitamin C solution. 5.00 cm³ of this vitamin C solution is added to a flask with approximately 150 cm³ of water and an indicator. Exactly 28.40 cm³ of 5.00 × 10⁻⁴ mol dm⁻³ I₂(aq) reacts with the sample of vitamin C solution in the flask.

[M_r : vitamin C, 176]

(iv) Suggest **two** reasons why hot concentrated acidified potassium manganate(VII) is **not** a suitable reagent for producing **Q** from vitamin C.

1

2

25. M/J 25/P22/Q3(a)(ii)-(iii)

Context: part (a) (introduction)

3 (a) C_3H_6 reacts with HBr(g) in an addition reaction.

- (ii) Complete Fig. 3.1 to show the mechanism for the addition reaction between C_3H_6 and HBr to produce 2-bromopropane. Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.

**Fig. 3.1**

- (iii) Explain why the major product of this reaction is 2-bromopropane rather than 1-bromopropane.

26. M/J 25/P22/Q3(e)

- (e) Alkene **Z** contains two $\text{C}=\text{C}$ bonds. **Z** reacts with an excess of hot concentrated acidified KMnO_4 to produce **only** CH_3COCH_3 , $\text{HOOCCH}_2\text{COOH}$, CO_2 and H_2O .

Suggest the structure of **Z**.

27. M/J 25/P23/Q3(a)(i)

Context: part (a) (introduction)

3 (a) Different hydrocarbon mixtures produced from fractional distillation of crude oil have different uses.

- (i) State the compound that is heated with long-chain hydrocarbons to produce more useful smaller alkanes and alkenes.

28. M/J 25/P23/Q3(b)(i)

Context: part (b) (introduction)

- (b) C_4H_8 reacts with an excess of H_2 to produce C_4H_{10} .



- (i) Name a catalyst for reaction 1.

29. M/J 25/P23/Q6(b)

- (b) Complete Fig. 6.1 to show the mechanism for the reaction between HOCl and $\text{CH}_2=\text{CH}_2$ to produce $\text{HOCH}_2\text{CH}_2\text{Cl}$.

Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.



Fig. 6.1

[4]

30. M/J 25/P24/Q5(d)

Context

- 5** The skeletal formula of **W** is shown in Fig. 5.1.

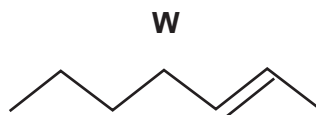


Fig. 5.1

- (d) **W** reacts with reagent **X** to produce ethanoic acid and pentanoic acid.

- (i) State the role of **X** in this reaction.

[1]

- (ii) Identify **X** and state the conditions used for this reaction.

[1]

31. MAR 25/P22/Q3(d)

also Unit 3 (1 mark), Unit 13 (1 mark); this unit 5 marks

Context

- 3** The halogens chlorine, bromine and iodine show trends in chemical and physical properties down the group.

Table 3.1 shows some properties of chlorine, bromine and iodine.

Table 3.1

property	chlorine	bromine	iodine
colour and state at room temperature	green gas		
bond energy / kJ mol^{-1}	242	193	151
electronegativity	3.0	2.8	2.5
formula of sodium halide	NaCl	NaBr	NaI

- (d)** Iodine monobromide, IBr , is a dark red solid that melts near room temperature.

IBr reacts with propene. The mechanism for this reaction is the same as the mechanism for that of HBr with propene.

- (i)** Identify **all** the intermolecular forces that exist between molecules of IBr .

[1]

- (ii)** Name the mechanism involved in the reaction of IBr with propene.

[1]

- (iii)** The reaction of IBr with propene forms two structural isomers.

Draw the **two** structural isomers shown by these molecules.



[2]

- (iv)** Identify the type of structural isomerism shown by the molecules in **(d)(iii)**.

[1]

- (v)** Explain why the two structural isomers do **not** form in equal amounts.

[2]

32. O/N 24/P21/Q2(b)(ii)-(iii)

Context

2 Chlorine is one of the elements in Group 17 of the Periodic Table.

Context: part (b) (introduction)

(b) The halogenoalkane $\text{CH}_3\text{CH}_2\text{Cl}$ forms when chlorine reacts with C_2H_6 via a free-radical substitution mechanism.**(ii)** State the essential condition for chlorine to react with C_2H_6 at room temperature.

[1]

(iii) Write **two** equations to show the propagation steps in this reaction.

1

2

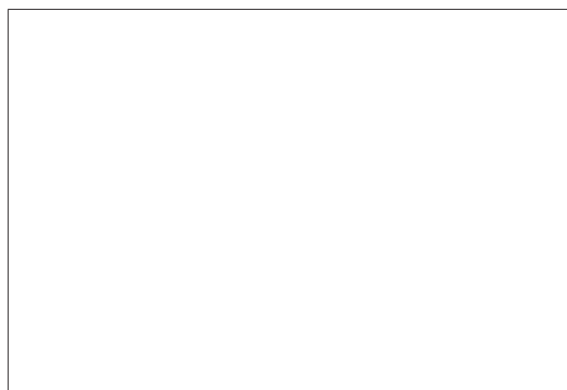
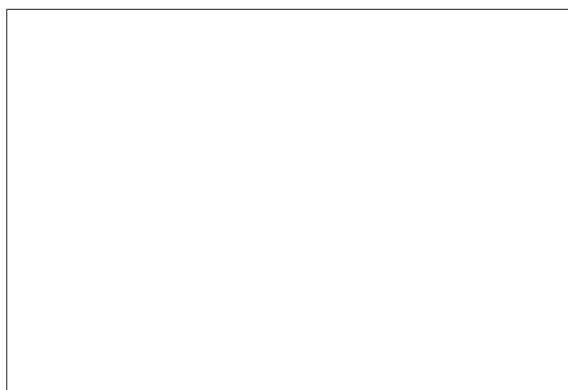
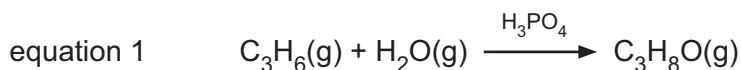
[2]

33. O/N 24/P22/Q2(b)(iii)

Context

2 Oxygen is a Group 16 element.

Context: part (b) (introduction)

(b) H_2O reacts with both inorganic and organic compounds.**(iii)** Draw the structures of the two alcohols formed in the reaction shown in equation 1.

[2]

34. O/N 24/P23/Q2(b)(ii)-(iii)

Context

2 Chlorine is one of the elements in Group 17 of the Periodic Table.

Context: part (b) (introduction)

(b) The halogenoalkane $\text{CH}_3\text{CH}_2\text{Cl}$ forms when chlorine reacts with C_2H_6 via a free-radical substitution mechanism.**(ii)** State the essential condition for chlorine to react with C_2H_6 at room temperature.

[1]

(iii) Write **two** equations to show the propagation steps in this reaction.

1

2

[2]

35. M/J 24/P21/Q5(c)(ii)

Context

5 Hydrocarbon molecules contain covalent bonds.

Context: part (c) (introduction)

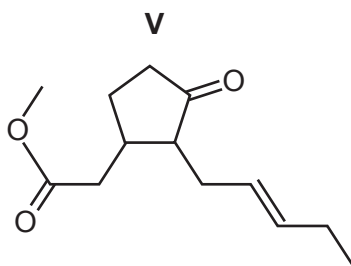
(c) In electrophilic reactions involving alkenes the π bond of $C=C$ is broken.**(ii)** Complete Fig. 5.1 to show the mechanism for the electrophilic addition of hydrogen bromide to 2-methylpropene to produce the major organic product.

Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.

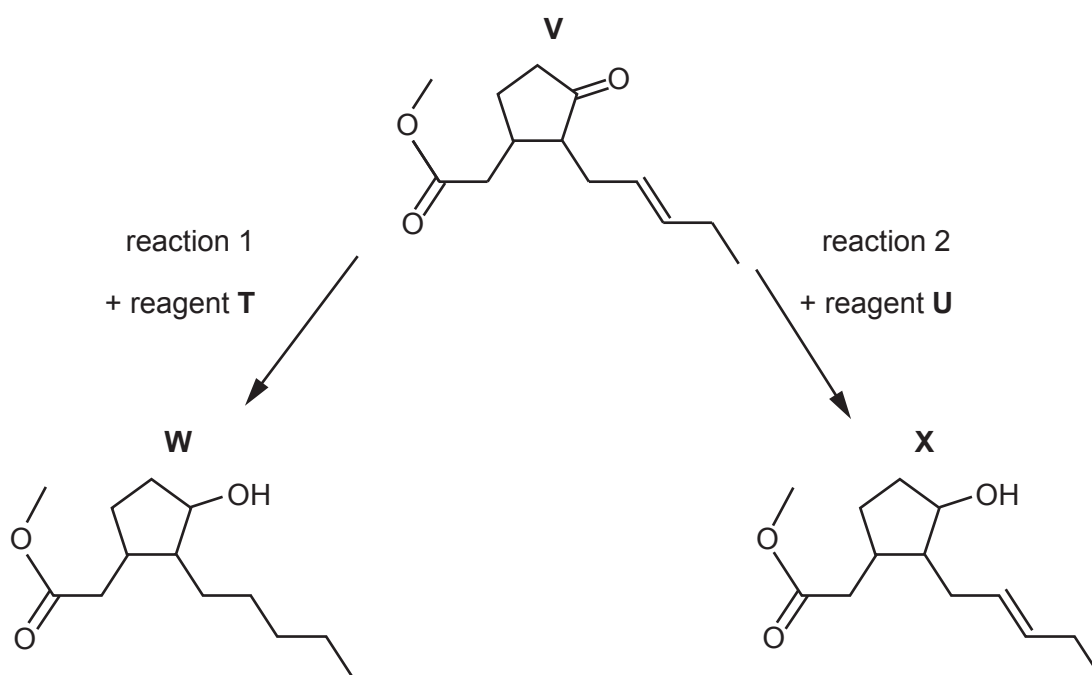
**Fig. 5.1****[4]**

36. M/J 24/P21/Q6(b)(ii)

Context: part (a) (introduction)

6 (a) V shows stereoisomerism.**Fig. 6.1**

Context: part (b) (introduction)

(b) Fig. 6.2 shows two reactions involving **V**.**Fig. 6.2****(ii)** Suggest the identity of reagent **U** in reaction 2.

37. M/J 24/P23/Q5(a)

5 (a) **M** reacts to form **R** by the addition of one reagent, as shown in Fig. 5.1.

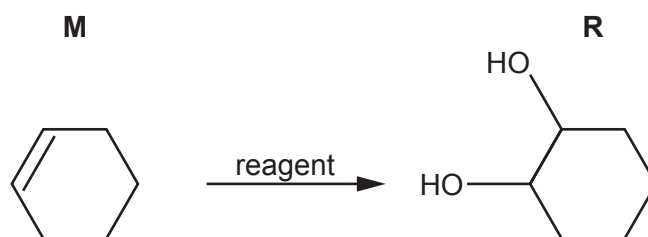


Fig. 5.1

Identify the reagent and conditions for this reaction.

[1]

38. M/J 24/P23/Q5(b)(ii)

Context: part (a)

5 (a) **M** reacts to form **R** by the addition of one reagent, as shown in Fig. 5.1.

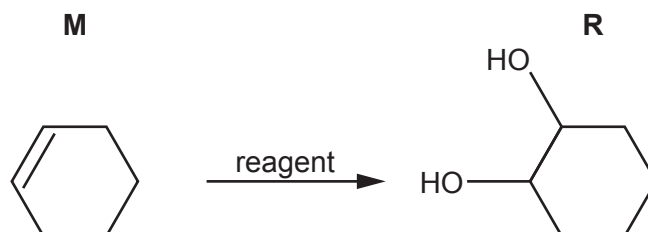


Fig. 5.1

Identify the reagent and conditions for this reaction.

[1]

Context: part (b) (introduction)

(b) **R** is also made from **M** by two steps, as shown in Fig. 5.2.

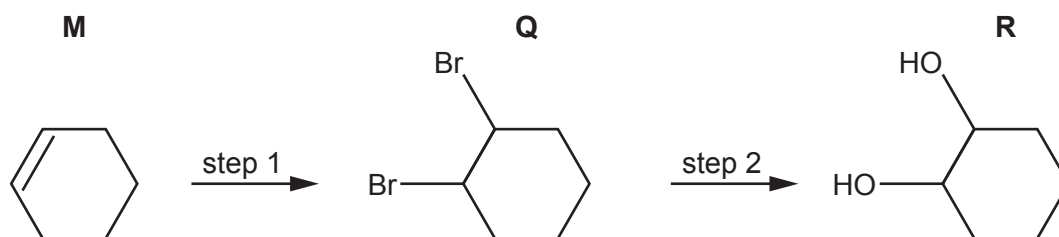


Fig. 5.2

(ii) Name the mechanism for step 1 in Fig. 5.2.

[1]

39. MAR 24/P22/Q1(d)(ii)

Context

- 1** Bismuth is an element in Group 15 of the Periodic Table.

Context: part (d) (introduction)

- (d)** NaBiO_3 is an oxidising agent with similar properties to KMnO_4 .

Fig. 1.1 shows an example of the use of NaBiO_3 as an oxidising agent.

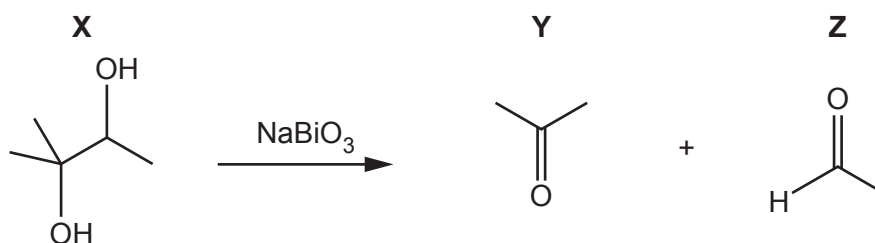


Fig. 1.1

- (ii)** Compound **X** forms when methylbut-2-ene reacts with KMnO_4 .

State the essential conditions for this reaction.

[1]

40. MAR 24/P22/Q2(d)(i)

Context

2 Chlorine, Cl_2 , reacts with many elements and compounds to form chlorides.

Table 2.1 shows information about some chlorides of Period 3 elements.

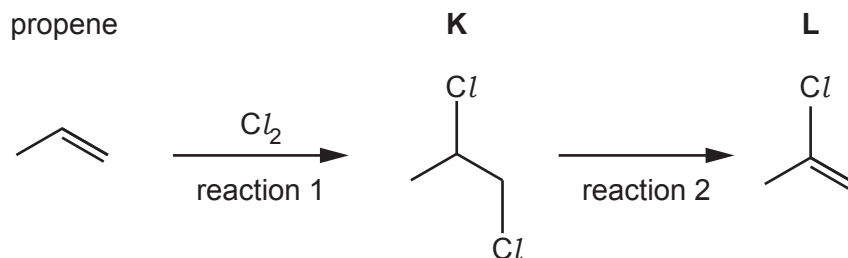
Table 2.1

	Na	Mg	Si
formula of chloride			
structure of chloride	giant		
bonding of chloride			covalent
pH of solution formed on addition of chloride to water		6.2	

Context: part (d) (introduction)

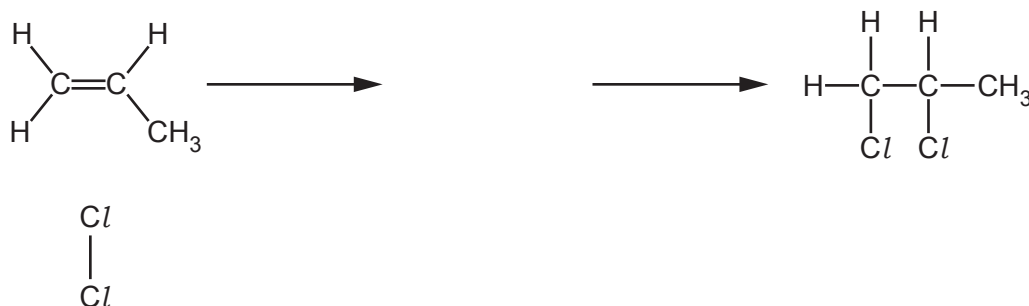
(d) Cl_2 reacts readily with propene to form **K**, 1,2-dichloropropane.

K can be used to form **L**.

**Fig. 2.1**

(i) Complete Fig. 2.2 to show the mechanism for the reaction of Cl_2 with propene in reaction 1.

Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.

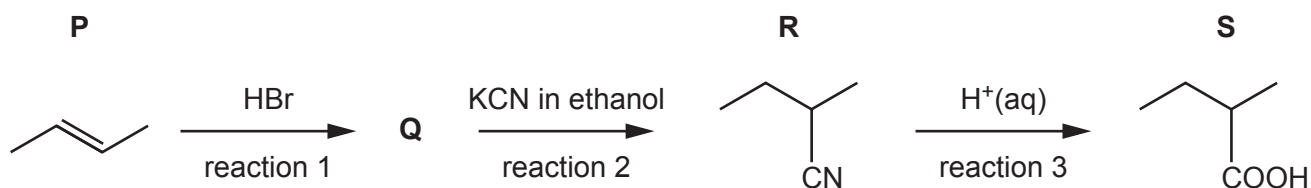
**Fig. 2.2**

[4]

41. MAR 24/P22/Q4(b)(i)

Context

4 Compound **S** is used in food flavourings. A possible synthesis of **S** is shown in Fig. 4.1.

**Fig. 4.1**

(b) (i) Give the structural formula of **Q**.

[1]

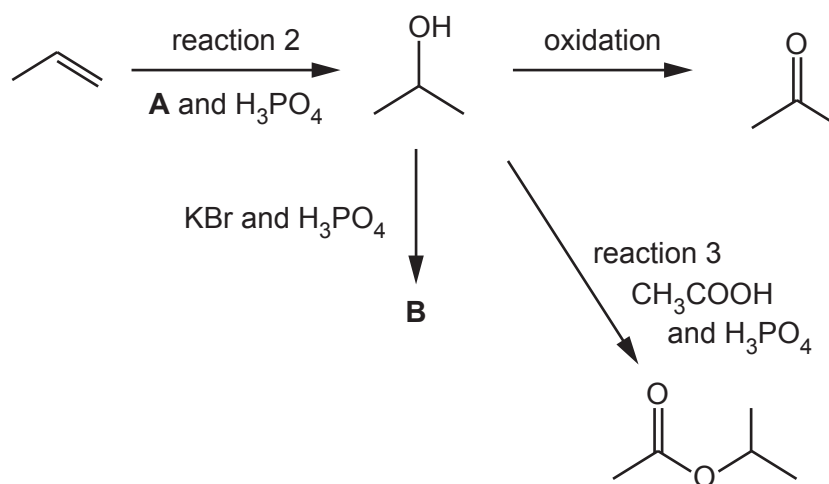
42. O/N 23/P22/Q3(c)(i)

Context

3 Phosphoric(V) acid, H_3PO_4 , is used in both inorganic and organic reactions.

Context: part (c) (introduction)

(c) Fig. 3.1 shows a reaction scheme that involves H_3PO_4 in several reactions.

**Fig. 3.1**

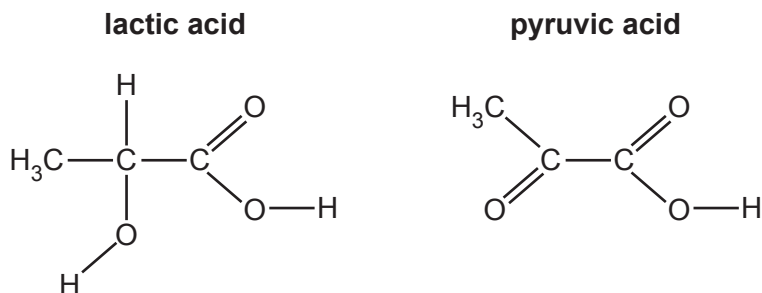
(i) Identify **A**, which reacts with propene in the presence of H_3PO_4 in reaction 2.

[1]

43. O/N 23/P22/Q4(b)(i)

Context

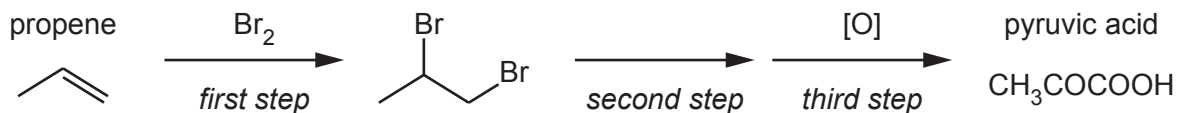
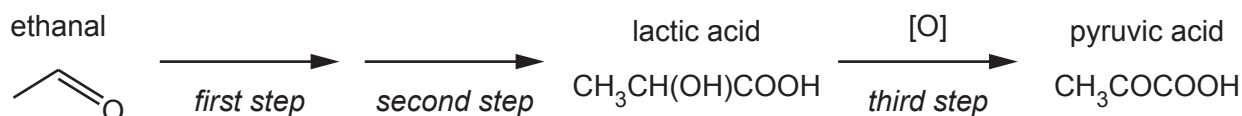
- 4 Lactic acid, $\text{CH}_3\text{CH}(\text{OH})\text{COOH}$, and pyruvic acid, $\text{CH}_3\text{COCO}\text{OH}$, both contain two functional groups.

**Fig. 4.1**

Context: part (b) (introduction)

- (b) Two possible syntheses of pyruvic acid are shown in Fig. 4.3 and Fig. 4.4.

Each synthesis has a total of three steps.

**Fig. 4.3****Fig. 4.4**

- (i) Complete the diagram in Fig. 4.5 to show the mechanism for the reaction of propene with Br_2 .

Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.

**Fig. 4.5**

44. M/J 23/P21/Q3(d)

Context

3 Chlorine is a very reactive element.**(d)** Chlorine reacts with methane in a series of reactions to produce chloroalkanes.**(i)** State the conditions required for chlorine to react with methane.

[1]

(ii) One of the products of the reaction is CH_2Cl_2 which reacts further to produce CHCl_3 .Complete Table 3.2 to show details of the mechanism that forms CHCl_3 from CH_2Cl_2 .**Table 3.2**

name of step	equation
initiation	
propagation	$\text{CH}_2\text{Cl}_2 + \text{Cl}\cdot \rightarrow$
termination	 $\rightarrow \text{CHCl}_3$

[3]

45. M/J 23/P21/Q5(a)**5 (a)** Molecule **M** is present in petrol, a fuel used in cars. **M** is a saturated, non-cyclic hydrocarbon. **M** contains eight carbon atoms.**(i)** Construct an equation for the complete combustion of **M**.

[2]

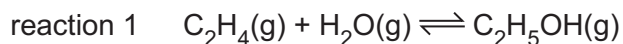
(ii) Describe how the composition of products differs when incomplete combustion of **M** occurs.

[2]

46. M/J 23/P22/Q4(c)(ii)-(iii)

Context

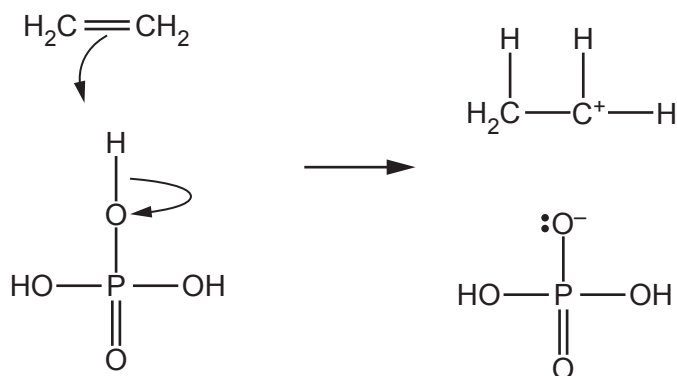
4 In industry, ethanol is made by reacting ethene with steam in the presence of H_3PO_4 .



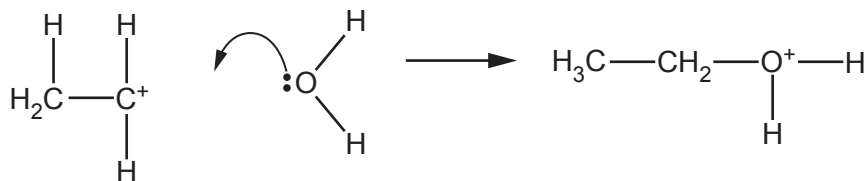
Context: part (c) (introduction)

(c) The mechanism for reaction 1 can be described in three steps. Steps 1 and 2 for reaction 1 are shown in Fig. 4.1.

step 1



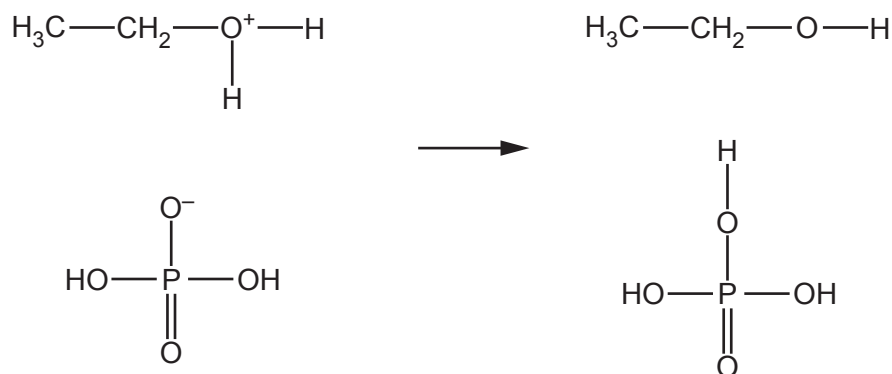
step 2

**Fig. 4.1**

(ii) Identify the species that behaves as an electrophile in step 2 in Fig. 4.1. Explain your answer.

[1]

(iii) Complete Fig. 4.2 to show the mechanism for step 3 of reaction 1. Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.

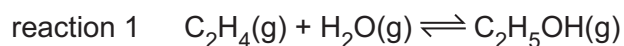
**Fig. 4.2**

[2]

47. M/J 23/P22/Q4(c)(vi)

Context

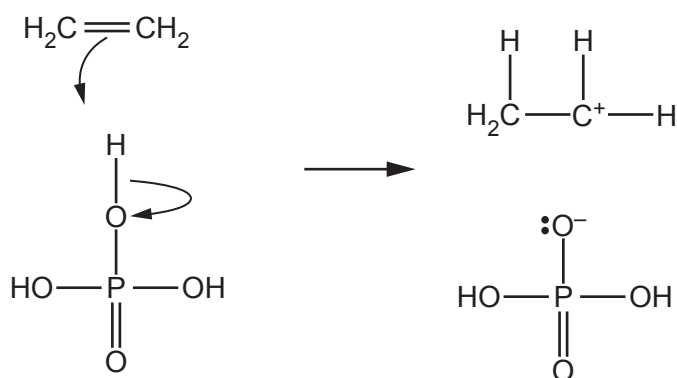
4 In industry, ethanol is made by reacting ethene with steam in the presence of H_3PO_4 .



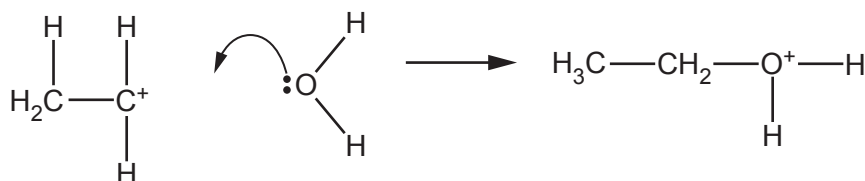
Context: part (c) (introduction)

(c) The mechanism for reaction 1 can be described in three steps. Steps 1 and 2 for reaction 1 are shown in Fig. 4.1.

step 1



step 2

**Fig. 4.1**

(vi) Propene also reacts with steam. A mixture of organic products is produced.

Explain why propan-2-ol is produced in the higher yield.

[2]

48. O/N 22/P21/Q5(a)(i)

Context

- 5** Lactones are cyclic esters. Under suitable conditions, lactones form from molecules that have both an alcohol and a carboxylic acid functional group.
Equation 1 shows an example of the formation of a lactone.

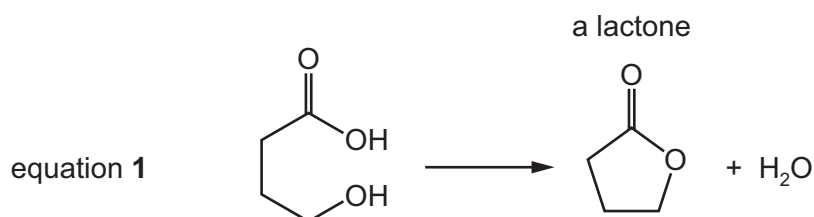


Fig. 5.1 shows the synthesis of lactone **P** from compound **M**.

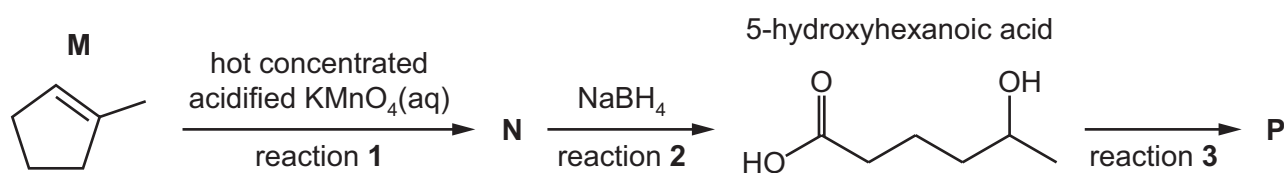


Fig. 5.1

- (a) (i)** **M** reacts with hot concentrated acidified KMnO₄(aq) to form **N**, C₆H₁₀O₃, in reaction 1.

Draw the structure of **N**.

[1]

49. O/N 22/P23/Q5(a)(i)

Context

- 5** Lactones are cyclic esters. Under suitable conditions, lactones form from molecules that have both an alcohol and a carboxylic acid functional group.
Equation 1 shows an example of the formation of a lactone.

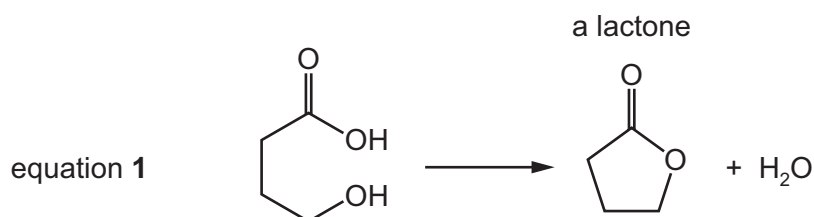


Fig. 5.1 shows the synthesis of lactone **P** from compound **M**.

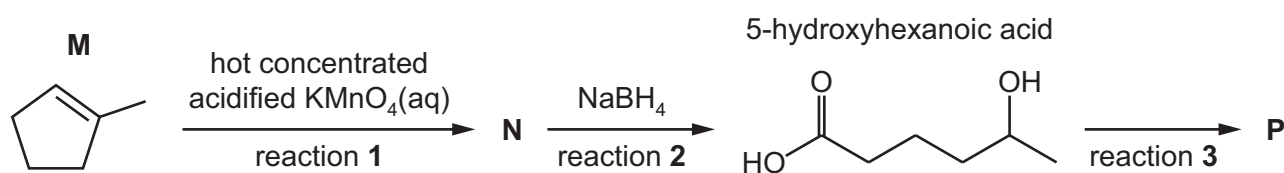


Fig. 5.1

- (a) (i)** **M** reacts with hot concentrated acidified $\text{KMnO}_4(\text{aq})$ to form **N**, $\text{C}_6\text{H}_{10}\text{O}_3$, in reaction 1.

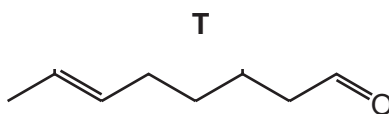
Draw the structure of **N**.

[1]

50. M/J 22/P21/Q3(b)

Context

- 3** Liquids that contain molecules of **T** smell like lemons.

**Fig. 3.1**

- (b)** Two organic products are produced when a sample of **T** is heated under reflux with excess acidified concentrated KMnO_4 .

Draw the structure of the two organic products, from this reaction, in the boxes.

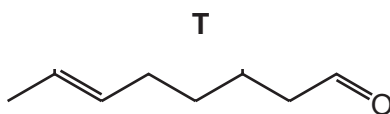


[2]

51. M/J 22/P21/Q3(c)(ii)-(iii)

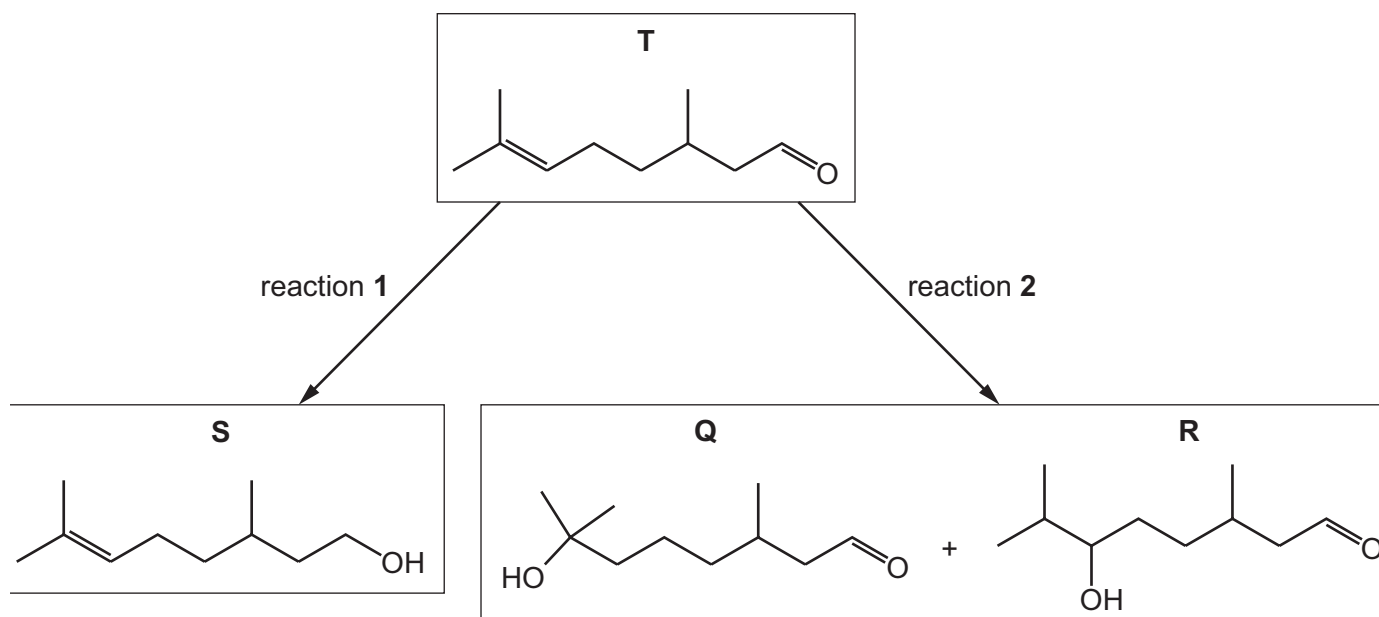
Context

- 3** Liquids that contain molecules of **T** smell like lemons.

**Fig. 3.1**

Context: part (c) (introduction)

- (c)** Fig. 3.2 shows two reactions of **T**.

**Fig. 3.2**

- (ii)** Identify the reagent and conditions needed for reaction 2.

[2]

- (iii)** Suggest which product formed in reaction 2 has a higher yield. Explain your answer.

[3]

52. M/J 22/P23/Q4(a)

- 4 (a) 2-methylpropene reacts with HCl(g) at room temperature. The major organic product is 2-chloro-2-methylpropane.

(i) Complete Fig. 4.1 to show the structure of the intermediate and mechanism for this reaction. Include charges, dipoles, lone pairs of electrons and curly arrows as appropriate.



Fig. 4.1

[3]

(ii) Explain why, in this reaction, 2-chloro-2-methylpropane is produced at a higher yield than 1-chloro-2-methylpropane.

[2]

53. M/J 22/P23/Q5(a)(i)

Context: part (a) (introduction)

- 5 (a) But-2-ene reacts with KMnO_4 to form organic product, **Y**.
Y does not react with Na_2CO_3 .
A gas is produced when an excess of Na is added to **Y**.

(i) Describe the conditions for the KMnO_4 used in the reaction to form **Y** from but-2-ene.

[1]

54. MAR 22/P22/Q3(e)(i)-(ii)

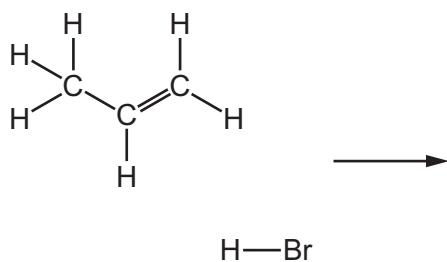
Context

3 The hydrogen halides HCl , HBr and HI are all colourless gases at room temperature.

Context: part (e) (introduction)

(e) HBr reacts with propene to form two bromoalkanes, $\text{CH}_3\text{CH}_2\text{CH}_2\text{Br}$ and $(\text{CH}_3)_2\text{CHBr}$.

- (i)** Complete the diagram to show the mechanism of the reaction of HBr and propene to form the major organic product.
Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.
Draw the structures of the intermediate and the major organic product.



[4]

- (ii)** Explain why the two bromoalkanes are **not** produced in equal amounts by this reaction.

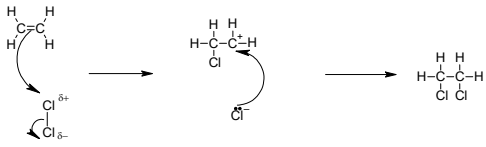
[2]

Answers Section

1. M/J 26/P21/Q5(b)

5(b)	electrophilic addition	1	1 addition alone
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2. M/J 26/P21/Q5(d)

5(d)	 M1 curly arrow from C=C bond to Cl closest to C=C M2 correct dipoles $Cl\delta^+ - Cl\delta^-$ AND curly arrow from bond of $Cl\delta^+ - Cl\delta^-$ to $Cl\delta^-$ M3 correct carbocation intermediate M4 curly arrow from lone pair of : Cl^- to C^+ of intermediate Judgement of arrows: A curly arrows that appear straight NOT single headed arrows (fishhooks) first time seen-and then A M1 judgement of arrows: π bonds M1 starts from anywhere on π bond AND goes to Cl or close to Cl atom of $Cl-Cl$ M1 if arrow does not start at the double bond, but when it is extrapolated back the base of the arrow hits the π bond AND the arrow end goes to Cl or close to Cl atom of $Cl-Cl$ M2 judgement of arrows: $Cl-Cl$ bond M2 A curly arrow from bond of $Cl-Cl$ to just beyond the Cl (if dipole correct).	4	M1 δ^-. I π or δ^- label on the double bond (e.g. $C = C$) I partial charges on Cl R partial charges on C of $C = C$ (e.g. $\delta^+C = \delta^+C / \delta^-C = \delta^-C$) R arrows from any other part of the alkene M2 A dipole moment of Cl_2 shown as an arrow with + convention I electrons (i.e. dots / crosses above and/or below bond) M3 A unambiguous structure R δ^+ charge on C^+ R lone pair on C^+ M4 A ecf from incorrect intermediate structure in M3 when no positive charge on C of intermediate OR $C\delta^+$ instead of C^+ I δ^-Cl^- if all else ok
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3. M/J 26/P22/Q5(b)(iii)-(iv)

5(b)(iii)	M1 $CH_3(CH_2)_4CH_3 + Cl\cdot \rightarrow CH_3(CH_2)_3\cdot CHCH_3 + HCl$ M2 $CH_3(CH_2)_3\cdot CHCH_3 + Cl_2 \rightarrow CH_3(CH_2)_3CHClCH_3 + Cl\cdot$	2	Independent marks A radicals with or without $\cdot$ M1 I alkyl radical as $\cdot C_6H_{13}$ OR $CH_3(CH_2)_5\cdot$ I position of the dot BOD $CH_3(CH_2)_3CH_3CH\cdot$ OR $CH_3(CH_2)_3CCH_3H\cdot$ M2 A ecf on their alkyl structure from M1 (even if M2 does not balance) M2 A alkyl radical as $\cdot C_6H_{13}$ OR $CH_3(CH_2)_5\cdot$ M2 ecf use of anion/cation
5(b)(iv)	M1 $CH_3(CH_2)_4CH_3 + \cdot Cl \rightarrow CH_3(CH_2)_4\cdot CH_2 + HCl$ M2 $CH_3(CH_2)_4\cdot CH_2 + CH_3(CH_2)_4\cdot CH_2 \rightarrow CH_3(CH_2)_{10}CH_3$ OR $2CH_3(CH_2)_4\cdot CH_2 \rightarrow CH_3(CH_2)_{10}CH_3$	2	Independent marks A radicals with or without $\cdot$ M1 A $CH_3(CH_2)_5\cdot$ M1 I alkyl radical as $\cdot C_6H_{13}$ I position of the dot M2 A alkyl radical as $\cdot C_6H_{13}$ OR $CH_3(CH_2)_5\cdot$ M2 ecf use of anion/cation

4. M/J 26/P22/Q5(c)

5(c)	number of alkyl / methyl groups	1	A number of C atoms bonded to the central carbon / branches / R groups BOD number of alkyl groups bonded to the carbocation I number of C atoms bonded to the carbocation I number of C atoms I number of ethyl/propyl groups I number of hydrogens I inductive effect I tertiary>secondary>primary stability I carbocation stability
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5. M/J 26/P22/Q6(b)(i)

6(b)(i)	M1 steam / H ₂ O (g) M2 H ₃ PO ₄ (catalyst) / phosphoric (V) acid (catalyst)	2	M1 A water <u>and</u> then temperature $\geq 100^\circ\text{C}$ (temperature might be seen on conditions line) M2 A H ₃ PO ₄ (catalyst) / phosphoric (V) acid (catalyst) / sulfuric acid (catalyst) / H ₂ SO ₄ (catalyst) on reagent line A orthophosphoric acid / phosphoric acid / sulfuric acid I heat under reflux / heat alone I pressure I state symbols
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6. M/J 26/P23/Q5(a)(ii)

5(a)(ii)	free-radical substitution	1	
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7. M/J 26/P23/Q5(b)(ii)-(iii)

5(b)(ii)	M1 $\text{CH}_3\text{CH}_2\text{CH}_3 + \bullet\text{Cl} \rightarrow \bullet\text{CH}_2\text{CH}_2\text{CH}_3 + \text{HCl}$ M2 $\bullet\text{CH}_2\text{CH}_2\text{CH}_3 + \text{Cl}_2 \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{Cl} + \bullet\text{Cl}$	2	M1 I $\bullet\text{C}_3\text{H}_7$ M1 R $\text{CH}_3\bullet\text{CHCH}_3$ I position of the dot I absence of the dot M2 A alkyl radical as $\bullet\text{C}_3\text{H}_7$ or $\text{CH}_3\bullet\text{CHCH}_3$
5(b)(iii)	M1 $\text{CH}_3\text{CH}_2\text{CH}_3 + \bullet\text{Cl} \rightarrow \text{CH}_3\bullet\text{CHCH}_3 + \text{HCl}$ M2 $2\text{CH}_3\bullet\text{CHCH}_3 \rightarrow (\text{CH}_3)_2\text{CHCH}(\text{CH}_3)_2$	2	A radicals with or without $\bullet$ $(\text{CH}_3)_2\text{CH}\bullet = \text{CH}_3\bullet\text{CHCH}_3$ M1 A C_3H_7 for CH_3CHCH_3 M1 R $\bullet\text{CH}_2\text{CH}_2\text{CH}_3$ I position of the dot M2 I alkyl radical as $\bullet\text{C}_3\text{H}_7$ M2 R $\bullet\text{CH}_2\text{CH}_2\text{CH}_3$

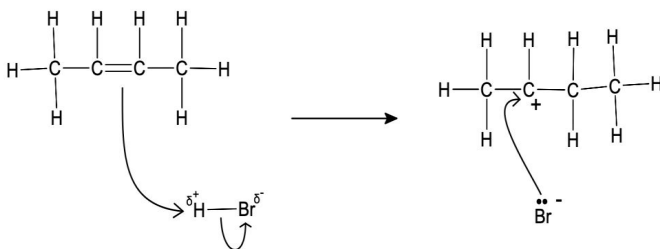
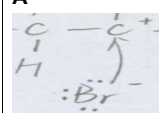
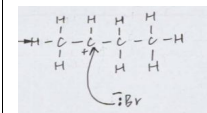
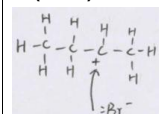
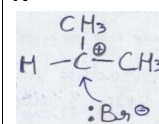
8. M/J 26/P24/Q5(a)

5(a)	Al ₂ O ₃ AND heat / high temperature	1	A zeolite / alumina / porous pot / porcelain / ceramics AND heat A high temperature/high temp = heat A $\geq 100^\circ\text{C}$ I catalyst I steam I pressure
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9. M/J 26/P24/Q5(b)(iii)

5(b)(iii)	orange / brown (solution / liquid) to colourless	1	I turns clear I colour disappears I discolouration / colour discharged A orange / brown disappears A decolourises / turns colourless A brown / yellow / yellow-brown / orange-brown / yellow-orange R red / red-brown I 'liquid'
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10. M/J 26/P24/Q5(d)

5(d)(i)	electron pair acceptor	1	I reference to likes electrons / attracted to electrons / area of low electron density. A lone pair acceptor A gains both electrons I electron acceptor
5(d)(ii)	 M1 curly arrow from $=$ of $C=C$ to H - I π or $\delta-$ label on the double bond - R partial charges shown on C of $C=C$ R arrows from any other part of the alkene M2 correct dipoles $H^{\delta+}-Br^{\delta-}$ AND curly arrow from bond of $H^{\delta+}-Br^{\delta-}$ to $Br^{\delta-}$ - I electrons (i.e. dots / crosses above and / or below bond) M3 correct intermediate – secondary carbocation only - A unambiguous structure - I $\delta+$ charge on C^+ - R lone pair on C^+	4	Judgement of arrows: A curly arrows that appear straight R single headed arrows first time seen and then A M1 judgement of arrows: π bonds M1 starts from anywhere on π bond AND goes to H or close to H atom of $H-Br$ ✓ M1 if arrow does not start at the double bond, but when it is extrapolated back the base of the arrow hits the π bond AND the arrow end goes to H or close to H atom of $H-Br$ ✓ M2 judgement of arrows: $H-Br$ bond M2 A curly arrow from bond of $H-Br$ to just beyond the Br (if dipole correct).
5(d)(ii) ctd.	M4 curly arrow from lone pair of $:Br^-$ to C^+ of intermediate A ecf from incorrect intermediate structure in M3 when: - no positive charge on C of intermediate OR $C^{\delta+}$ instead of C^+ - I $\delta-$ on $:Br^-$		Judgement of arrows: lone pairs M4 A arrow starts anywhere closely associated with lone pair A  A  A (BOD)  R 

11. MAR 26/P22/Q3(c)

3(c)(i)	(free-)radical substitution	1
3(c)(ii)	M1 <i>propagation</i> $\text{Cl}^\bullet + \text{C}_4\text{H}_{10} \rightarrow \text{C}_4\text{H}_9^\bullet + \text{HCl}$ M2 <i>termination</i> $\text{C}_4\text{H}_9^\bullet + \text{Cl}^\bullet \rightarrow \text{C}_4\text{H}_9\text{Cl}$	2

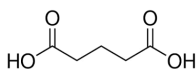
12. O/N 25/P21/Q3(c)(ii)-(iii)

3(c)(ii)	M1 $\text{C}_5\text{H}_{10} + \text{Cl}^\bullet \rightarrow \text{C}_5\text{H}_9^\bullet + \text{HCl}$ M2 $\text{C}_5\text{H}_9^\bullet + \text{Cl}_2 \rightarrow \text{C}_5\text{H}_9\text{Cl} + \text{Cl}^\bullet$	2
3(c)(iii)	termination	1

13. O/N 25/P21/Q3(d)(i)

3(d)(i)	HCl (g)	1
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14. O/N 25/P21/Q3(e)(i)

3(e)(i)		1
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15. O/N 25/P22/Q3(b)

3(b)	M1 strong C—H (bonds) M2 non-polar (C—H / molecule / bonds)	2
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16. O/N 25/P22/Q3(c)

3(c)	cracking	1
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17. O/N 25/P22/Q3(e)(i)

3(e)(i)	H_2 ((g)) OR hydrogen (gas)	1
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18. O/N 25/P22/Q3(e)(iii)

3(e)(iii)	$\text{C}_6\text{H}_{14} + 9\frac{1}{2}\text{O}_2 \rightarrow 6\text{CO}_2 + 7\text{H}_2\text{O}$	1
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19. O/N 25/P22/Q4(a)(i)-(ii)

4(a)(i)	free-radical substitution	1
4(a)(ii)	M1 $\text{C}_3\text{H}_8 + \text{Cl}^\bullet \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2^\bullet + \text{HCl}$ OR $\text{C}_3\text{H}_8 + \text{Cl}^\bullet \rightarrow \text{C}_3\text{H}_7^\bullet + \text{HCl}$ M2 $\text{CH}_3\text{CH}_2\text{CH}_2^\bullet + \text{Cl}_2 \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{Cl} + \text{Cl}^\bullet$ OR $\text{C}_3\text{H}_7^\bullet + \text{Cl}_2 \rightarrow \text{C}_3\text{H}_7\text{Cl} + \text{Cl}^\bullet$ etc M3 $\text{CH}_3\text{CH}_2\text{CH}_2^\bullet + \text{Cl}^\bullet \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{Cl}$ OR $\text{C}_3\text{H}_7^\bullet + \text{Cl}^\bullet \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{Cl}$	3

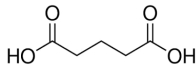
20. O/N 25/P23/Q3(c)(ii)-(iii)

3(c)(ii)	M1 $\text{C}_5\text{H}_{10} + \text{Cl}^\bullet \rightarrow \text{C}_5\text{H}_9^\bullet + \text{HCl}$ M2 $\text{C}_5\text{H}_9^\bullet + \text{Cl}_2 \rightarrow \text{C}_5\text{H}_9\text{Cl} + \text{Cl}^\bullet$	2
3(c)(iii)	termination	1

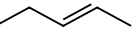
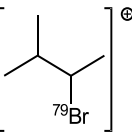
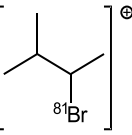
21. O/N 25/P23/Q3(d)(i)

3(d)(i)	HCl (g)	1
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22. O/N 25/P23/Q3(e)(i)

3(e)(i)		1
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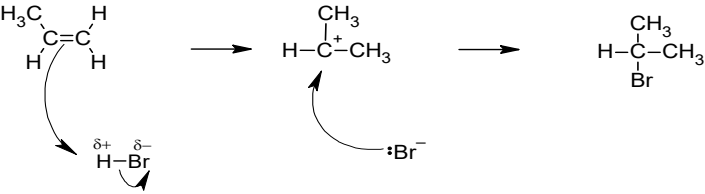
23. O/N 25/P24/Q4(a)

4(a)(i)	M1 (molecules with the) same molecular formula / same number of atoms of each element M2 different structural formulae/structures	2
4(a)(ii)	M1  M2 restricted / lack of / no rotation about C=C	2
4(a)(iii)	electrophilic addition	1
4(a)(iv)	M1 CH ₃ COCH ₃ + CH ₃ COOH as products M2 (CH ₃) ₂ C = CHCH ₃ + 3[O] → CH ₃ COCH ₃ + CH ₃ COOH	2
4(a)(v)	M1 (reaction proceeds via a) more stable intermediate / carbocation M2 because of greater positive inductive effect of more alkyl groups	2
4(a)(vi)	M1  M2 	2

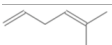
24. M/J 25/P21/Q4(b)(iv)

4(b)(iv)	M1 (also) oxidises / reacts with (primary and / or secondary) alcohol groups M2 (also) breaks the sigma and pi bond in C=C	2
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25. M/J 25/P22/Q3(a)(ii)-(iii)

3(a)(ii)	 M1 curly arrow from = of C=C to H (of HBr) M2 curly arrow from bond of H ^{δ+} —Br ^{δ-} to Br ^{δ-} (dipole is essential) M3 correct intermediate M4 curly arrow from lone pair of Br ⁻ to C ⁺ of intermediate drawn	4
3(a)(iii)	M1 (2-bromopropane is made from the) more stable intermediate / carbocation M2 (due to positive) inductive effect of more alkyl groups (attached to the C ⁺ / central carbon atom) OR (due to) greater (positive) inductive effect of 2 methyl groups (rather than 1 ethyl group)	2

26. M/J 25/P22/Q3(e)

3(e)	$\text{H}_2\text{C}=\text{CHCH}_2\text{CH}=\text{C}(\text{CH}_3)_2$ ✓✓  ✓✓ $\text{H}_2\text{CCHCH}_2\text{CHC}(\text{CH}_3)_2$ ✓✓ M1 $\text{H}_2\text{C}=\text{C}\dots$ on one end of their molecule OR $\dots\text{C}=\text{C}(\text{CH}_3)_2$ on one end of their molecule OR $\dots\text{C}=\text{CHCH}_2\text{CH}=\text{C}\dots$ in the middle of their molecule M2 correct structure	2
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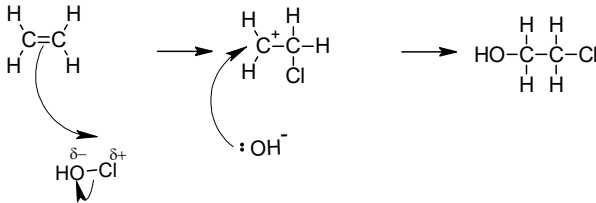
27. M/J 25/P23/Q3(a)(i)

3(a)(i)	Al_2O_3 / aluminium oxide	1
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28. M/J 25/P23/Q3(b)(i)

3(b)(i)	nickel / platinum	1
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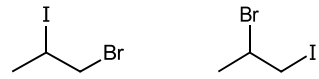
29. M/J 25/P23/Q6(b)

6(b)	 M1 correct arrow for $\text{HO}-\text{Cl}$ M2 arrow from double bond of $\text{C}=\text{C}$ to Cl M3 correct intermediate M4 arrow from lone pair on O of $:\text{OH}^-$ to the appropriate C	4
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30. M/J 25/P24/Q5(d)

5(d)(i)	oxidising agent	1
5(d)(ii)	hot concentrated acidified KMnO_4	1

31. MAR 25/P22/Q3(d)

3(d)(i)	instantaneous dipole—induced dipole / id—id / London / dispersion (forces) AND permanent dipole—permanent dipole / pd—pd (forces)	1
3(d)(ii)	electrophilic addition	1
3(d)(iii)		2
3(d)(iv)	position(al) / regio- (isomerism)	1
3(d)(v)	- secondary (carbo)(cat)ion/intermediate is more stable / $[\text{CH}_3\text{CHCH}_2\text{I}]^+$ is preferred - electrophile approaches with I nearest / $\text{I}^{\delta+}$ / I^+ attacks first - I and Br have different electronegativities / IBr is a polar molecule	2

32. O/N 24/P21/Q2(b)(ii)-(iii)

2(b)(ii)	ultraviolet	1
2(b)(iii)	$\text{C}_2\text{H}_6 + \text{Cl}^\bullet \rightarrow \text{C}_2\text{H}_5^\bullet + \text{HCl}$	1
	$\text{C}_2\text{H}_5^\bullet + \text{Cl}_2 \rightarrow \text{C}_2\text{H}_5\text{Cl} + \text{Cl}^\bullet$	1

33. O/N 24/P22/Q2(b)(iii)

2(b)(iii)		2
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34. O/N 24/P23/Q2(b)(ii)-(iii)

2(b)(ii)	ultraviolet	1
2(b)(iii)	$C_2H_6 + Cl^\bullet \rightarrow C_2H_5^\bullet + HCl$	1
	$C_2H_5^\bullet + Cl_2 \rightarrow C_2H_5Cl + Cl^\bullet$	1

35. M/J 24/P21/Q5(c)(ii)

5(c)(ii)	M1 arrow from = of C=C to H of H-Br M2 correct dipole on H-Br AND arrow from H-Br bond to Br M3 correct intermediate AND 2-bromopropane M4 arrow from lone pair on Br⁻ to C⁺ of their intermediate (with +ve or δ+ charge)	4
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36. M/J 24/P21/Q6(b)(ii)

6(b)(ii)	sodium borohydride / NaBH ₄	1
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37. M/J 24/P23/Q5(a)

5(a)	cold dilute KMnO ₄ OR cold dilute potassium manganate(VII)	1
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38. M/J 24/P23/Q5(b)(ii)

5(b)(ii)	electrophilic addition	1
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Answer for context part (a)

5(a)	cold dilute KMnO ₄ OR cold dilute potassium manganate(VII)	1
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39. MAR 24/P22/Q1(d)(ii)

1(d)(ii)	(KMnO ₄ is) cold and dilute	1
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40. MAR 24/P22/Q2(d)(i)

2(d)(i)	• curly arrow from C=C bond to Cl(δ⁺) • correct dipole on Cl₂ AND curly arrow from Cl-Cl bond to Cl^{δ-} • correct carbocation intermediate • curly arrow from lone pair on Cl⁻ to C⁺	1
		1
		1
		1

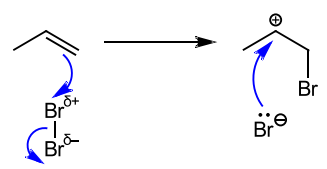
41. MAR 24/P22/Q4(b)(i)

4(b)(i)	molecule	P	Q	R	S	1
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42. O/N 23/P22/Q3(c)(i)

3(c)(i)	H ₂ O / steam	1
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43. O/N 23/P22/Q4(b)(i)

4(b)(i)	 curly arrow from C=C bond to Br^{δ+} AND curly arrow from Br^{δ+}—Br^{δ-} bond to Br^{δ-} AND Br^{δ+} closest to the double bond correct intermediate curly arrow from lone pair on Br⁻ to C⁽⁺⁾ of the intermediate	1
		1
		1

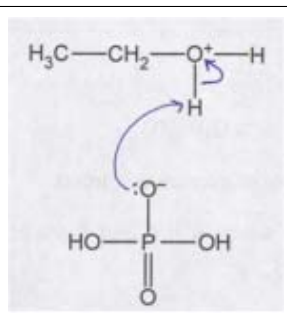
44. M/J 23/P21/Q3(d)

3(d)(i)	ultraviolet light			1	
3(d)(ii)				3	
		name of stage	equation		
		initiation	$\text{Cl}_2 \rightarrow 2\text{Cl}\cdot$		[1]
		propagation	$\text{CH}_2\text{Cl}_2 + \text{Cl}\cdot \rightarrow \text{CHCl}_2\cdot + \text{HCl}$		[1]
		termination	$\text{CHCl}_2\cdot + \text{Cl}\cdot \rightarrow \text{CHCl}_3$		[1]

45. M/J 23/P21/Q5(a)

5(a)(i)	$\text{C}_8\text{H}_{18} + 12\frac{1}{2} \text{O}_2 \rightarrow 8\text{CO}_2 + 9\text{H}_2\text{O}$ M1 correct formula of M C ₈ H ₁₈ M2 balanced equation showing complete combustion of M	2
5(a)(ii)	• C • CO • unburnt hydrocarbons • less/no CO₂ Two correct for one mark, three correct for two marks.	2

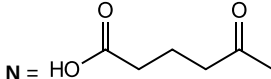
46. M/J 23/P22/Q4(c)(ii)-(iii)

4(c)(ii)	(CH ₃ CH ₂) ⁺ OR CH ₃ C ⁺ H ₂ AND accepts a pair of electrons	1
4(c)(iii)	 M1 arrow from bond of one of the O-H to O of the cation M2 arrow from lone pair on O of the anion to between H of H-O	2

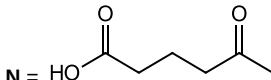
47. M/J 23/P22/Q4(c)(vi)

4(c)(vi)	M1 (propan-2-ol is produced via a) more stable carbocation OR more stable intermediate OR $(\text{CH}_3)_2\text{C}^+\text{H}$ is more stable M2 (positive) inductive effect of more alkyl/methyl groups OR more electron donating alkyl / methyl groups OR greater electron donating effect of two alkyl / two methyl groups	2
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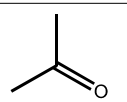
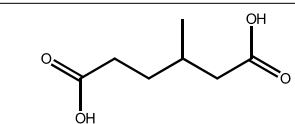
48. O/N 22/P21/Q5(a)(i)

5(a)(i)		1
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49. O/N 22/P23/Q5(a)(i)

5(a)(i)		1
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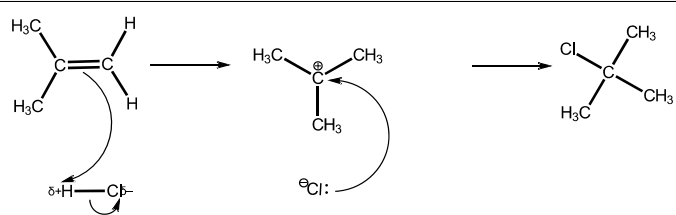
50. M/J 22/P21/Q3(b)

3(b)	 OR $(\text{CH}_3)_2\text{CO}$  OR $\text{HOOC}(\text{CH}_2)_2\text{CH}(\text{CH}_3)\text{CH}_2\text{COOH}$	2
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51. M/J 22/P21/Q3(c)(ii)-(iii)

3(c)(ii)	M1 $\text{H}_2\text{O}(\text{g})$ / steam M2 phosphoric acid (catalyst) / H_3PO_4 (catalyst)	2
3(c)(iii)	M1 Q AND intermediate / cation is (more) stable M2 (as the) C^+ has 3 / more alkyl groups attached M3 (so) greater (positive) inductive effect	3

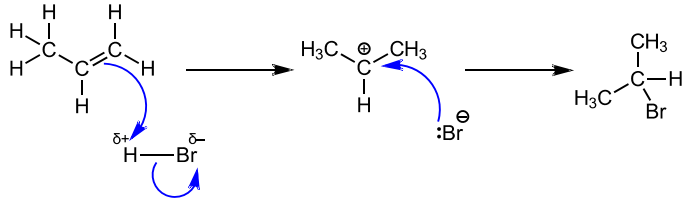
52. M/J 22/P23/Q4(a)

4(a)(i)	 M1 curly arrow from = to H AND correct dipole on H-Cl M2 correct intermediate M3 curly arrow from H-Cl bond to Cl AND curly arrow from lp on: Cl^- to C^+	3
4(a)(ii)	- intermediate OR C^+ OR carbocation (formed) is more stable - due to greater (positive) inductive effect. - caused by 3 methyl groups / 3 alkyl groups / more alkyl groups attached to C^+ / central C	2

53. M/J 22/P23/Q5(a)(i)

5(a)(i)	cold + dilute	1
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54. MAR 22/P22/Q3(e)(i)-(ii)

3(e)(i)	 M1: curly arrow from C=C to H of HBr M2: correct dipole ($\delta^+ \text{H}-\text{Br}^{\delta-}$) AND curly arrow from H—Br to Br M3: curly arrow from lone pair on :Br⁻ to carbocation M4: correct intermediate AND product (2-bromopropane)	4
3(e)(ii)	2° carbocation more stable (than 1°) (or reverse argument) [1] greater positive inductive effect of two methyl groups (<i>cf.</i> one ethyl) (or reverse argument) [1]	2